Sequences and Basic Series Convergence Tests

ANSWER KEY

Section 1: Limits of Sequences & Monotonicity

1 Determine the limit of the sequence $a_n = \frac{4n^2 - 3n}{2n^2 + 5}$.

a. 2 ✓

b. 4

c. ∞ (diverges)

d. 0

Divide numerator and denominator by n^2: $\lim_{n\to\infty} \frac{4 - 3/n}{2 + 5/n^2} = \frac{4}{2} = 2$.

2 Which statement is guaranteed by the Monotone Convergence Theorem for Sequences?

a. **Every bounded monotonic sequence converges** ✓

b. Every alternating sequence converges

c. Every bounded sequence is monotonic

d. Every convergent sequence is positive

The Monotone Convergence Theorem states that if a sequence is bounded and monotonic (either non-decreasing or non-increasing), it must converge.

3 What is $\lim_{n\to\infty} (1 + \frac{2}{n})^n$?

a. e^2 ✓

b. e

c. 1

d. ∞

By the standard exponential limit $\lim_{n\to\infty} (1 + \frac{x}{n})^n = e^x$, with $x = 2$, the limit is e^2.

Section 2: Geometric Series & n-th Term Divergence Test

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4 Evaluate the sum of the series $\sum_{n=1}^{\infty} 3(\frac{2}{5})^n$.

a. 2 ✓

b. $\frac{6}{5}$

c. $\frac{15}{2}$

d. $\frac{5}{2}$

First term $a = 3(\frac{2}{5}) = \frac{6}{5}$, common ratio $r = \frac{2}{5}$. Sum $= \frac{a}{1-r} = \frac{6/5}{1-2/5} = \frac{6/5}{3/5} = 2$.

5 If $\lim_{n\to\infty} a_n = \frac{1}{2}$, what does the n-th Term Test conclude about $\sum_{n=1}^{\infty} a_n$?

a. **The series diverges** ✓

b. The series converges to $\frac{1}{2}$

c. The test is inconclusive

d. The series is geometric

The n-th Term Divergence Test states that if $\lim_{n\to\infty} a_n \neq 0$, the series must diverge.

6 If $\lim_{n\to\infty} a_n = 0$, what can be concluded about $\sum a_n$?

a. **The test is inconclusive; the series may converge or diverge** ✓

b. The series converges

c. The series diverges

d. The sum is zero

Having terms approach zero is a necessary condition for convergence, but not sufficient (e.g., the harmonic series $\sum \frac{1}{n}$ has terms approaching 0 but diverges).

Section 3: Integral Test & p-Series

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7 For what values of p does the p-series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ converge?

a. $p > 1$ ✓

b. $p \geq 1$

c. $p < 1$

d. $p > 0$

By the p-series test, $\sum \frac{1}{n^p}$ converges if and only if $p > 1$. For $p \leq 1$, it diverges.

8 Which series converges by the Integral Test?

a. $\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^2}$ ✓

b. $\sum_{n=2}^{\infty} \frac{1}{n \ln n}$

c. $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$

d. $\sum_{n=1}^{\infty} \frac{1}{2n+1}$

$\int_2^{\infty} \frac{1}{x(\ln x)^2} dx = [-\frac{1}{\ln x}]_2^{\infty} = \frac{1}{\ln 2} < \infty$. Since the improper integral converges, the series converges.

9 What three conditions must $f(x)$ satisfy on $[1, \infty)$ to apply the Integral Test to $\sum a_n$?

a. $f(x)$ **must be continuous, positive, and decreasing** ✓

b. $f(x)$ must be differentiable, alternating, and bounded

c. $f(x)$ must be positive and increasing

d. $f(x)$ must be a rational function

The hypotheses of the Integral Test require $f(x)$ to be continuous, positive, and decreasing on $[1, \infty)$ where $f(n) = a_n$.

Section 4: Direct Comparison & Limit Comparison Tests

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10 Determine the convergence of $\sum_{n=1}^{\infty} \frac{1}{n^3+5n}$.

a. **Converges by Direct Comparison with $\sum \frac{1}{n^3}$** ✓

b. Diverges by Comparison with $\sum \frac{1}{n}$

c. Diverges by n-th term test

d. Inconclusive

For all $n \geq 1$: $\frac{1}{n^3+5n} < \frac{1}{n^3}$. Since $\sum \frac{1}{n^3}$ is a convergent p-series ($p = 3 > 1$), the given series converges by Direct Comparison.

11 Which series should be used for the Limit Comparison Test with $\sum_{n=1}^{\infty} \frac{3n+2}{2n^3-1}$?

a. $\sum \frac{1}{n^2}$ **(convergent)** ✓

b. $\sum \frac{1}{n}$ (divergent)

c. $\sum \frac{1}{n^3}$ (convergent)

d. $\sum n$ (divergent)

The dominant terms are $\frac{3n}{2n^3} = \frac{3}{2n^2}$. Comparing with $b_n = \frac{1}{n^2}$ yields a positive finite limit $\frac{3}{2}$, confirming convergence.

12 In the Limit Comparison Test, if $\lim_{n\to\infty} \frac{a_n}{b_n} = L$ where $0 < L < \infty$, what does this mean?

a. **Both $\sum a_n$ and $\sum b_n$ either both converge or both diverge** ✓

b. $\sum a_n$ converges to L

c. $\sum a_n$ converges while $\sum b_n$ diverges

d. The test is inconclusive

A finite positive limit L ensures that both series share the exact same convergence or divergence behavior.

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